

Demonstration of Proposed Features

Date	6 July2026
Team ID	
Project Name	Plugging into the Future : An Exploration of Electricity Consumption Patterns
Maximum Marks	1Mark

Demonstration of Proposed Features:

S.No	Feature Name	Description	Status (Implemented/ Partial / Pending)	Demonstrate d(Yes/No)	Remarks
1	Power Consumption Data Input	Collects regional and seasonal electricity usage data via the web form.	Implemented	Yes	Working correctly
2	Data Preprocessing	Cleans and transforms raw time-series consumption data for model prediction.	Implemented	Yes	Successfully tested
3	Consumption Visualization	Displays dataset patterns using time-series graphs, count plots, and distribution trends.	Implemented	Yes	Charts generated
4	Machine Learning Model Training	Trains forecasting models using the historic electricity load dataset.	Implemented	Yes	Models trained
5	Electricity Demand Prediction	Predicts future consumption patterns, load demands, and peak usage hours.	Implemented	Yes	Accurate prediction
6	Analytics Dashboard Display	Displays the final demand forecasting trends and metrics on the Flask web page.	Implemented	Yes	Result Displayed successfully

7	User Authentication	Allows users to securely log in before accessing the forecasting system dashboard.	Implemented	Yes	Login Working successfully
8	Dashboard Analytics	Displays total consumption metrics, seasonal peak demands, and model prediction confidence.	Implemented	Yes	Dashboard functioning correctly
9	Demand Forecasting Prediction	Displays the confidence percentage or error margin of each electricity load prediction.	Implemented	Yes	Prediction metrics displayed correctly
10	SQLite Database Integration	Stores historical power consumption logs and past prediction records in the SQLite database.	Implemented	Yes	Data saved successfully

Feature Implementation Summary:

ITEM	VALUE
Total Features Proposed	10
Total Features Implemented	10
Total Features Demonstrated	10
Overall ImplementationRate(%)	100%